

A Framework for Emergent Time from Horn-Filling Dynamics

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Abstract

We propose a framework in which spacetime is a growing simplicial set and time is the process of filling its inner horns. Two local, non-deleting moves drive the growth of spacetime: a "sprout" attaches a fresh directed edge ending at a new vertex, and a "fill" completes an unfilled inner horn with a composite edge and its witnessing cell. Each move is a weak homotopy equivalence of the underlying space, so the homotopy type is conserved at every step. Since the geometric realization of the simplicial set forgets edge orientation, this conserved type cannot see any arrow of time; all physical content lives in the directed structure that realization forgets. The completed limit of the filling process is a quasicategory, in which every composition question has an answer while edges remain directed. Each move is an event that consumes cells created by earlier events, the dependency of moves forms a partial order, and a reference frame is a choice of linearization of this order. The entropy of an event is the logarithm of the number of linear extensions of its causal past. This quantity measures how much of the past could have happened in a different order, and grows monotonically, which we prove; a toy implementation provides first numerical support. The arrow of time appears as a broken symmetry. The universe carries an arrow exactly when it is not isomorphic to its own edge-reversal, and pairing the universe with its mirror image restores the symmetry globally, in the spirit of CPT-symmetric cosmologies. We formulate a maximum-entropy inference problem whose solution, conjecturally of Gibbs form, would determine the rule from locality and a fixed update budget alone. We close with the open problems, foremost the recovery of the Lorentz group and of an area law for horizon entropy.

1 Introduction

Discrete approaches to quantum gravity, from Regge calculus [22] to causal dynamical triangulations [3], replace smooth spacetime with combinatorial data: simplices, gluings, and the lengths assigned to them. In Regge calculus the combinatorial scaffold is fixed with only edge lengths allowed to vary. The vertices, simplices, and gluings of the triangulation never change, so no new combinatorial structure is ever introduced. That is enough for classical gravity, where one solves the field equations on a triangulation that is given in advance. The discomfort begins one step later, when Feynman's path integral is applied to the evolution of the universe and the sum runs over all histories. Summing over Regge geometries means integrating over edge lengths, a continuous space in which many assignments describe the same geometry and no canonical measure presents itself. On the other hand, dynamical triangulations [11, 1, 2] escape this by inverting Regge's choice. The lengths are frozen and the combinatorics is set free, so a geometry becomes a gluing pattern of simplices whose shapes are fixed once and for all, and summing over geometries becomes counting gluings.

In both cases the geometry is carried by the gluing and not by any ambient space. Each simplex is flat inside, so curvature can only sit where simplices meet, and it is measured by how far the angles around such a meeting place fall short of a full turn. Figure 1 shows the three cases for triangles glued around a vertex.

Because the combinatorics is now free, the number of simplices differs from history to history, and within a single history the number of simplices in a slice changes from one slice to the next.

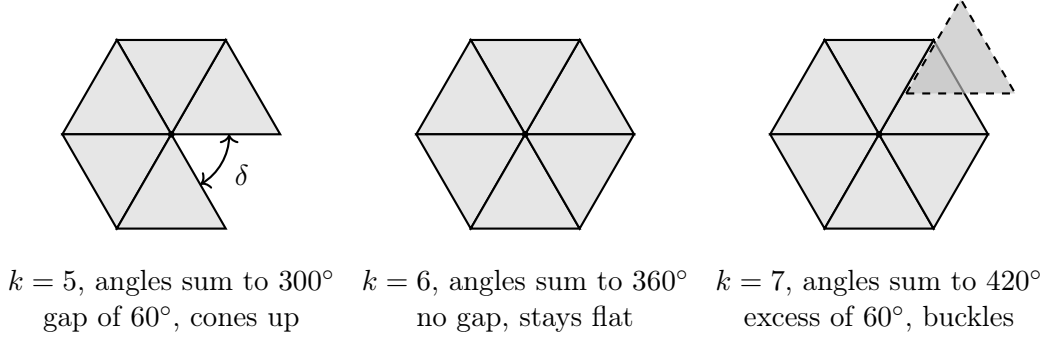


Figure 1: Flat equilateral triangles glued around a single vertex. With five triangles the angles at the vertex sum to 300° , the surface cannot lie flat, and it cones up. With six they sum to 360° and the surface is genuinely flat. With seven they sum to 420° , there is no room for the seventh triangle in the plane, and the surface buckles into a saddle. The shortfall $\delta = 360^\circ - \sum_i \theta_i$ is called the deficit angle, and it accounts for all of the curvature of the glued surface.

New geometric measure appears from nowhere. Classical general relativity is no better off on this point, since it has no conservation law for spatial volume either, and in a closed universe the Hamiltonian constraint balances the total at zero [4, 12] however much volume appears. The complaint is therefore not that volume grows, but that the growth answers to nothing the theory conserves. A cosmological constant term only assigns a cost to four-volume within the ensemble, and where the total is held fixed instead, as in simulations and in unimodular gravity [26, 15], the constraint is imposed by hand rather than descending from a symmetry.

This paper takes the discomfort seriously and asks what a theory looks like if one demands, from the start, that

1. Nothing is ever destroyed. Evolution proceeds only by forking and completion moves, never by deletion, so that topological invariants are conserved by construction.
2. Time is not a dimension of the fundamental structure but an emergent bookkeeping of an irreversible forking-filling process.
3. Entropy is attached to frame-invariant causal data, not to spatial slices, so that the theory never needs to say what all of space looks like at a single moment.
4. The rule that chooses which move happens next is derived from an inference principle rather than postulated.

In our framework the universe begins as a single vertex, which we call the big bang, and grows by two kinds of local moves:

1. A **sprout** attaches a fresh directed edge ending at a new vertex, adding new space.
2. A **fill** completes an unfilled inner horn, attaching a composite edge together with the higher cell that witnesses the composition.

Two kinds of direction appear here. Every edge of the simplicial set carries an orientation, and that orientation is what decides which horns count as inner and what composition means. Separately, every move is an event, and the events are ordered by which of them created the cells the others consumed. We call the first the **orientation** and the second the **causal order**. No move ever deletes anything, so growth always runs forward in the causal order, while orientation is a property of the finished cells and can be reversed on paper.

At the big bang two edges sprout from the single vertex with opposite orientation, and this is what produces the two branches. The mirror branch is not an independent run. By definition it is the edge-reversal of the branch, carrying one mirrored event for every event, which is what makes the pair a mirror image of itself. The mirroring is causally inert, since no move in one branch ever consumes a cell created in the other, so the causal past of any event lies wholly inside its own branch. Each branch grows forward in its own causal order while pointing the opposite way in orientation, so neither branch runs backward in time in any absolute sense. Each is backward only as seen from the other, and there is no global causal order in which either of them is reversed. This is a combinatorial counterpart of CPT-symmetric cosmologies, where a universe and an antiuniverse emerge together from a single point [7, 5].

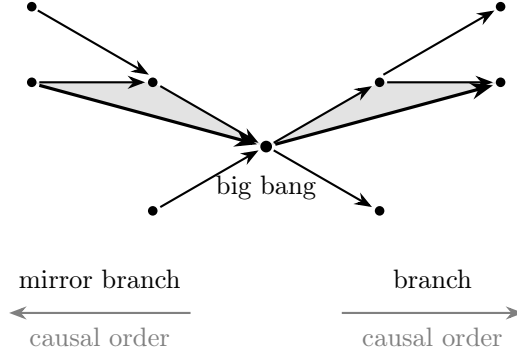


Figure 2: The total object. A branch on the right and its mirror image on the left, glued at the single vertex where both were born. Corresponding edges carry opposite orientation, so the pair is isomorphic to its own edge-reversal. The shaded region on each side is a filled inner horn, drawn with its composite edge in bold, and the fill is mirrored along with everything else. The grey arrows are the causal order, which runs away from the shared vertex on both sides, so each branch is backward only as seen from the other, and never in the order in which its own cells were built.

At any moment the state of the universe is a finite simplicial set with directed edges, together with the history of how it was built. The limit of the filling process in one branch is a quasicategory, in which every composition has been formed while the orientations remain. The classical, zero-information limit is the nerve of an ordinary category, the simplicial set whose n -cells are the chains of n composable morphisms [24, 19], and in it every filler is unique. The physical content of a completed universe is precisely how far it is from being a nerve.

The pair of branches is meant to carry no information at all. Two contractible halves glued at a single vertex are again contractible, so every homotopy invariant of the pair is that of a point, and the pair is isomorphic to its own edge-reversal, so it carries no arrow either. Whatever is physical therefore sits inside a single branch, in the orientations and in the history of how its cells were built.

2 Preliminaries and related work

2.1 Simplicial sets, horns, and quasicategories

Definition 2.1 (Simplicial set). Let Δ denote the category whose objects are the finite nonempty linear orders $[n] = \{0 < 1 < \dots < n\}$ and whose morphisms are order-preserving maps. A *simplicial set* is a functor $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$. The set $X_n = X([n])$ is the set of n -simplices; X_0 are *vertices*, X_1 are (directed) *edges*. The category of simplicial sets is denoted $\mathbf{sSet}$.

Example 2.2. The *standard n -simplex* $\Delta^n = \Delta(-, [n])$. Concretely, Δ^2 has three vertices $0, 1, 2$, three edges $0 \rightarrow 1, 1 \rightarrow 2, 0 \rightarrow 2$, and one nondegenerate 2-simplex filling the triangle.

Definition 2.3 (Horn). For $0 \leq k \leq n$, the *horn* $\Lambda_k^n \subset \Delta^n$ is the subcomplex obtained by removing the interior of Δ^n and the face opposite vertex k . A horn is *inner* if $0 < k < n$.

Example 2.4. Λ_1^2 consists of two composable edges $0 \rightarrow 1 \rightarrow 2$ with the composite $0 \rightarrow 2$ and the filling triangle *missing*. Filling Λ_1^2 means supplying both.

Definition 2.5 (Kan complex; quasicategory; nerve). A simplicial set X is a *Kan complex* if every map $\Lambda_k^n \rightarrow X$ extends along $\Lambda_k^n \hookrightarrow \Delta^n$ for all horns, and a *quasicategory* if this holds for all *inner* horns. The *nerve* $N(\mathcal{C})$ of a small category $\mathcal{C}$ is the simplicial set with $N(\mathcal{C})_n =$ chains of n composable morphisms. A simplicial set is a nerve of a category iff every inner horn has a *unique* filler [19].

Remark 2.6 (The spectrum used in this paper). Bare directed graphs (only 0- and 1-cells): nothing filled. Quasicategories: everything filled, fillers unique only up to higher cells. Nerves: everything filled uniquely. We will interpret these respectively as the newborn universe, the completed universe, and the classical (zero-information) limit.

Remark 2.7 (Opposites). The involution of Δ reversing each linear order induces $X \mapsto X^{\text{op}}$, under which Λ_k^n becomes Λ_{n-k}^n . The two outer horns are exchanged and inner horns go to inner horns, so X is a quasicategory exactly when X^{op} is. Reversing orientations therefore cannot turn an inner-horn filling policy into an outer-horn one.

2.2 Rewriting, events, and causal order

Definition 2.8 (Non-deleting rewriting rule). A *rule* is a monomorphism $L \hookrightarrow R$ of finite simplicial sets. An *application* of the rule to a configuration X is a choice of match $m : L \rightarrow X$; the result is the pushout $X \cup_L R$. Because $L \hookrightarrow R$ is a monomorphism and nothing is removed, applications only add cells. (This is the non-deleting special case of double-pushout graph rewriting [13], where the interface equals the left-hand side.)

Definition 2.9 (Run; event; causal order). A *run* is a finite sequence of rule applications starting from an initial object. Each application is an *event*. Event b *depends* on event a (written $a \prec b$) if the match of b uses a cell created by a ; the reflexive-transitive closure $\leq$ of $\prec$ is the *causal order*. The *causal past* of an event e is $J^-(e) = \{a : a \leq e\}$.

Remark 2.10 (Orientation is not the causal order). Two directions must be kept apart. The *orientation* of an edge is part of the simplicial set and decides which horns are inner. The causal order of Definition 2.9 is an order on events and records which move supplied the cells another move consumed. Since no rule deletes anything, the causal order is a partial order whatever the orientations do, and a branch whose edges all point towards its origin still grows away from it.

Remark 2.11. The triple (events, causality, independence) is an event structure in the sense of Winskel [27]; we use only the partial order.

Definition 2.12 (Linearization; frame). A *linearization* (or *linear extension*) of a finite poset $(P, \leq)$ is a total order on P extending $\leq$. We write $\text{Ext}(P)$ for the number of linear extensions. In this framework a *reference frame* is a linearization of the causal order.

2.3 Associahedra and the coherence tower

Definition 2.13 (Catalan numbers; associahedron). The n -th Catalan number is $C_n = \binom{2n}{n} / (n+1)$; C_{n-1} counts the full bracketings of n composable edges. The *associahedron* K_n [25] is the polytope whose vertices are these bracketings and whose faces encode partial bracketings; K_4 is the pentagon.

The relevance is dynamical: filling 2-horns creates composites, whose alternative bracketings pose 3-horn questions, whose answers pose compatibility (4-horn, pentagon) questions, and so on. Completion promotes questions up this tower; it never depletes them. Since the number of bracketings grows like $C_{n-1} \sim 4^n / n^{3/2}$, question supply grows exponentially with tower level.

2.4 Related work

[GAP: This subsection must be revised after the adversarial literature scan is complete. Overlap verdicts below are provisional.]

Discrete gravity. Regge [22] showed that curvature on a piecewise-flat complex is concentrated on hinges as deficit angles and that the corresponding action converges to the Einstein–Hilbert action. Dynamical triangulations freeze edge lengths and make the gluing combinatorics dynamical; causal dynamical triangulations [3] additionally impose a global time foliation and recover a de Sitter-like volume profile numerically. Our framework shares the combinatorial ontology but differs in three respects: moves are non-deleting (CDT’s Monte Carlo moves are not), no foliation is imposed (causal order is derived from move dependency), and the object grown is a directed simplicial set headed toward a quasicategory rather than a triangulated pseudomanifold of fixed topology.

Causal sets. Causal set theory [6] takes a locally finite partial order as the fundamental ontology, with order-plus-number determining geometry, and classical sequential growth [23] supplies a stochastic birth process. Our causal order is likewise fundamental-but-derived (from move dependency), and our sprouting move is a birth event; the essential difference is that our elements are not featureless points but cells of a simplicial set, so that composition, coherence, and the entropy of Section 3.3 have combinatorial substrate. To our knowledge the linear-extension entropy of a causal past has not been used as a dynamical functional in the causal set literature. [GAP: verify this claim in the lit scan]

Rewriting cosmologies. The Wolfram physics project [14] evolves hypergraphs by rewriting rules and argues that *causal invariance* (confluence of the rewriting system) yields observer-independence of the causal graph and, conjecturally, Lorentz covariance. We adopt the same event/causal-graph architecture and the same open problem (Section 5.2); our divergences are the specific simplicial move set (horn filling, with the quasicategory as limit object), the derived rather than postulated rule (Section 3.6), and the entropy functional.

Emergent and thermal time. That time may be state-dependent is the thermal time hypothesis of Connes and Rovelli [10], built on Tomita–Takesaki modular flow; that time may be entanglement is Page and Wootters [21]. Our proposal is coarser but of the same family: time is identified with an irreversible completion process, and equilibrium (the quasicategory limit, no unfilled inner horns) is the state in which the process halts.

CPT-symmetric cosmology. Pairing a universe with its mirror image so that the total object carries no arrow is the combinatorial counterpart of the universe-antiuniverse pair of Boyle et al. [7] and of the two arrows emerging from a shared middle in the Janus point picture of Barbour et al. [5]. In both of those the mirror branch is determined by the branch rather than evolving independently, which is also our convention.

Entropic derivations. Jaynes’ maximum-entropy inference [17], its dynamical extension by Caticha [9], and Jacobson’s derivation of the Einstein equations from the Clausius relation on local causal horizons [16] together define the strategy of Sections 3.6 and 5.2: derive the rule by inference, and aim the long game at an area law.

Computation-budget kinematics. The Margolus–Levitin bound [20] caps the rate of distinguishable state changes by energy, and Lloyd [18] applies it cosmologically. Our budget axiom (Section 3.5) is of this lineage.

3 The framework

3.1 Axioms

Axiom 1 (Ontology). The fundamental object is the run, a finite sequence of non-deleting moves starting from a single vertex. The state of the universe at an event is the colimit of that run’s causal past (Definition 3.5), so it is determined by the run rather than being independent data.

Remark 3.1 (The object is pre-geometric). The simplicial set is not spacetime and it is not a representation of a spacetime lying elsewhere. It carries no metric, no dimension and no manifold structure, and everything geometric has to be recovered from it (Section 3.4 and problem O4). One consequence constrains any such recovery. Coherence questions escalate without bound, so a run creates cells of arbitrarily high dimension. Higher cells cannot therefore be directions one can move along, and geometry must be built from the one-skeleton together with the causal order, with cells above dimension one acting as witnesses that a composition holds and contributing nothing metric.

Axiom 2 (Dynamics). Evolution proceeds by two local, non-deleting moves: *sprout* and *fill* (Definition 3.3). Only inner horns may be filled.

Axiom 3 (Budget). The rate of moves is fixed. The constant c is the exchange rate between translation and internal filling (Section 3.5).

Remark 3.2 (What the budget axiom leaves open). A rate requires a clock, and time in this framework is emergent, a frame being only a linearization of the causal order. Read as a universal number of moves per tick, the axiom would reinstate a preferred linearization. The intended reading is the energetic one of Margolus and Levitin [20], in which the budget bounds distinguishable changes per unit energy rather than per unit of any global time, and it awaits a notion of energy internal to the framework. The axiom is also silent on whether the budget is one number for the whole universe or one number per region. Locality favours the latter, and the two readings give different cosmologies (Section 3.7).

Axiom 4 (Inference). The probability distribution over available moves is the maximum-entropy distribution consistent with Axioms 2–3 and locality (Section 3.6). No further dynamical input is allowed.

3.2 State space and moves

Definition 3.3 (Moves). Fix the two rules, as monomorphisms of finite simplicial sets:

1. **Sprout:** $\Delta^0 \hookrightarrow \Delta^1$. A match is a vertex v ; the application attaches a fresh directed edge from v to a new vertex.
2. **Fill:** $\Lambda_k^n \hookrightarrow \Delta^n$ for $0 < k < n$. A match is an unfilled inner horn; the application attaches the missing face and interior. Each horn may be filled at most once per distinct match.
[GAP: decide and justify the at-most-once convention versus allowing multiple fillers of the same horn; interacts with the choice-entropy variant of Section 5.1]

The initial object is a single vertex Δ^0 (the big-bang event).

Remark 3.4 (Conservation for free). Sprouting is a Whitehead elementary expansion and inner-horn filling is an anodyne extension; both are homotopy equivalences. Hence the classical homotopy type of the universe is conserved by construction; indeed, it remains that of a point. All evolving information is therefore *directed* structure, invisible to classical homotopy invariants. This is a feature: the framework’s physics is located strictly below classical homotopy equivalence, in composition and coherence data.

Definition 3.5 (State as colimit of the causal past). Let e be an event of a run, with causal past $J^-(e)$ (Definition 2.9). The *state at e* is the colimit of the diagram of rule applications in $J^-(e)$. Since colimits depend only on the diagram and not on any ordering of it, the state at e is independent of the choice of linearization: no foliation enters.

Proposition 3.6 (Well-definedness). *The dependency relation of Definition 2.9 is a partial order, and the colimit in Definition 3.5 exists and is independent of linearization.*

[GAP: Proof to be written (target: next session). Sketch: antisymmetry from acyclicity of creation-before-use; colimit exists since sSet is cocomplete; independence is the universal property.]

3.3 The entropy functional

Definition 3.7 (Causal-past entropy). For an event e of a run,

$$S(e) = \log \text{Ext}(J^-(e)),$$

the logarithm of the number of linear extensions of its causal past.

The coarse-graining implicit in S is canonical to the framework: the state (colimit) forgets the order of causally independent events, and S counts exactly the forgotten orderings. A chain has $S = 0$; an antichain of n events has $S = \log n!$. Entropy measures causal parallelism.

Lemma 3.8 (Monotonicity). *If $e \leq e'$ then $S(e) \leq S(e')$.*

[GAP: Proof to be written (owner: ixaxaar; the injection of extension sets should be a half-page). Empirically verified in Section 4.]

Lemma 3.9 (Foliation invariance). *$S(e)$ depends only on the isomorphism class of the poset $J^-(e)$ and in particular is independent of any linearization of the run.*

[GAP: Proof to be written; content is the frame-independence of $J^-(e)$.]

Remark 3.10 (Cost of computation). Counting linear extensions is #P-complete in general [8]; Section 4 uses exact counting on small runs and future work requires a sampling estimator.

3.4 Space, distance, and speed

The axioms so far say nothing about space, and no statement about speed has content until they do. This section fixes the minimum, in order.

Definition 3.11 (Weighted one-skeleton). Every edge created by a sprout has weight 1. Every composite edge created by a fill has weight equal to the total weight of the path it composes. The *distance* between two vertices is the least total weight of an undirected path between them in the one-skeleton.

Remark 3.12 (Fills are metrically inert). A composite edge joins two vertices that were already joined by a path of exactly that weight, so under Definition 3.11 a fill changes no distance and creates no reachability that was not already present. The convention is forced if distance is to be stable. Weighting every composite by 1 instead makes each fill a shortcut, and accumulated shortcuts collapse the diameter of the complex in the manner of a small-world graph, leaving neither a stable distance function nor a bounded signal speed. The price of Definition 3.11 is that fills contribute nothing geometric. What they contribute is coherence data, which is where this framework locates its physical content.

Definition 3.13 (Track; translation; internal evolution). A *track* is a chain in the causal order whose consecutive events are matched at cells within bounded distance of one another. An event on a track is a *translation* if it is a sprout, extending the track's locus by one unit of distance, and *internal* if it is a fill, leaving the locus where it is.

Proposition 3.14 (A maximal signal speed). *Along any track, distance advances by at most one unit per event, so the speed of propagation of causal influence is bounded by one unit of distance per event. We write c for this bound.*

[GAP: Proof to be written. Content: by Remark 3.12 a fill adds no reachability, and a sprout adds one vertex at distance 1 from its match.]

The bound of Proposition 3.14 is not derived from anything deep. It is the statement that the moves are local, and c is a unit rather than a prediction. What is not settled, and is the content of problem O1, is why the resulting light-cone structure should be isotropic and the same for every observer.

3.5 Kinematics: proper time and mass

Definition 3.15 (Proper time). The proper time between two causally related events is the length of the longest chain in the causal order between them.

This is the causal set definition of discrete proper time. It is frame-independent, since the longest chain is a property of the poset alone. A null track, all of whose events are translations, has proper time zero, which recovers the statement that a photon carries no internal clock, and conversely a system with internal evolution accumulates proper time and is massive. Mass is the rest-frame rate of internal fills, in the spirit of the Compton clock $mc^2/\hbar$.

Remark 3.16 (Why longest chains, and what dilation still costs). Counting a track's own events gives the wrong dilation law. If a track has N events of which k are internal, the budget splits linearly and the internal rate is $1 - v/c$, where relativity requires $\sqrt{1 - v^2/c^2}$. The two disagree by a factor of three at $v = 0.8c$. The quadratic form comes from the Euclidean geometry of the light-clock argument rather than from any bookkeeping over moves, which is why Appendix A states that composition as an assumption. Definition 3.15 is chosen because it is the definition for which the quadratic form is a theorem. For a sprinkling into flat spacetime the longest chain between two events is proportional to $\sqrt{t^2 - x^2}$, and the straight route maximises it. What that theorem assumes is that the causal order resembles a sprinkling into a Lorentzian manifold, so dilation in this framework rests on problem O1 and is not established independently of it.

One further constraint the framework must respect. Uniform motion has to be locally undetectable, so all internal processes of a moving subsystem must slow together and rate differences may only appear on reunion. Any locally measurable change of internal rate under uniform motion would falsify the framework at once.

3.6 The maximum-entropy problem

Definition 3.17 (Move ensemble). At a given state, let $M = M_{\text{sprout}} \sqcup M_{\text{fill}}$ be the set of available moves (legal sprout sites and unfilled inner horns), and let $k : M \rightarrow \mathbb{R}_{>0}$ be a cost function depending only on a move's level d in the coherence tower.

Definition 3.18 (The inference problem). Find the probability distribution q on M maximizing the Shannon entropy $H(q) = -\sum_m q(m) \log q(m)$ subject to

(C1) normalization, $\sum_m q(m) = 1$;

(C2) budget, $\mathbb{E}_q[k] = R$;

(C3) locality and homogeneity: $q(m)$ depends only on the isomorphism class of the local neighbourhood of m 's match.

Conjecture 3.19 (Gibbs rule). *The problem of Definition 3.18 has a unique solution of the form $q(m) \propto e^{-\lambda k(m)}$ with a single multiplier $\lambda(R)$. The sprout-to-fill ratio and the expansion law $N(t)$ are then determined by the level degeneracies (Catalan counts) and k , with no further choices.*

[GAP: Derivation is the subject of a dedicated session. Danger to watch: whether (C3) can be formulated without smuggling in a preferred scale.]

3.7 Coherence escalation and the growth law

Filling promotes questions up the associahedral tower (Section 2). A fill at level d mints questions at level $d + 1$, with supply growing like 4^d , so completion never depletes the supply of available moves and no region ever finishes.

The tower taxes fills alone. Sprouts have constant cost and remain available at every vertex, and by Remark 3.12 fills are metrically inert, so the growth of space is carried by sprouts and coherence escalation exerts no direct brake on it. The tower acts only through competition for budget, and under a Gibbs rule expensive moves lose share, so the mix shifts towards sprouting as the typical level rises.

The growth law is therefore decided by the second question of Remark 3.2. With one budget for the whole universe the move rate saturates at a constant once cheap sprouts dominate, vertex count grows linearly, and the scale factor decelerates. With a budget per region the total activity is proportional to the number of vertices, growth is exponential, and the late universe is a de Sitter phase. Locality favours the second reading. We assert neither, and we draw no comparison with the observed history of deceleration followed by late-time acceleration, since Remark 3.1 and problem O4 stand between vertex count and any scale factor. **[GAP: Settle the global versus local budget question and derive the growth law for at least one explicit cost function k ; compare against the toy simulation once the sampling estimator allows long runs.]**

Remark 3.20 (Two arrows). The framework contains two independent monotone quantities: the entropy S (Lemma 3.8) and the tower level d . That they are aligned is expected but unproved. **[GAP: prove or refute alignment]**

4 Numerical results

We implemented the dynamics of Definition 3.3 as a stochastic process on directed graphs (code in Appendix B; the implementation currently realizes sprouts and Λ_1^2 fills, i.e. tower level $d \leq 2$). At each tick a sprout occurs with probability p_{sprout} , otherwise a uniformly random unfilled inner horn is filled. Dependencies follow Definition 2.9: a sprout at v depends on the event that created v ; a fill of the horn ($u \rightarrow v \rightarrow w$) depends on the events that created its two edges. The entropy of Definition 3.7 is computed exactly by dynamic programming over downsets of the event poset, which limits runs to ≈ 20 events (the #P wall).

Three observations, with their status:

1. **Monotonicity** (supports Lemma 3.8): $S(t)$ is strictly increasing in every run.
2. **Ratio ordering**: $S(t)$ is pointwise increasing in p_{sprout} . Interpretation: expansion produces entropy; equilibration (filling) suppresses its growth. This links the entropy functional to the expansion dynamics and is the empirical hook for treating the dynamics as an entropic gradient flow. **[GAP: significance analysis: more seeds, error bars, and a test of pointwise ordering]**

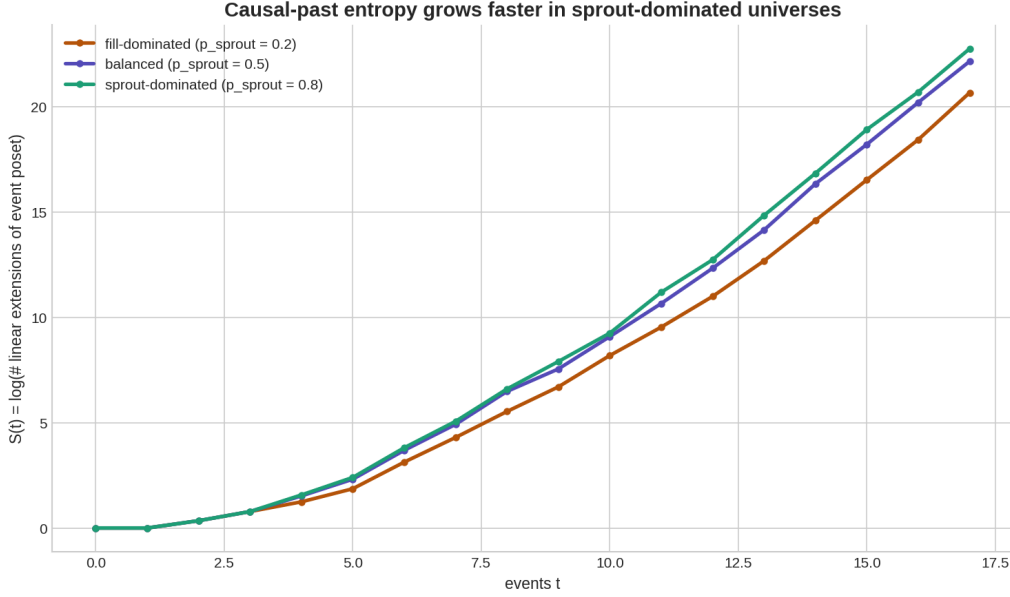


Figure 3: Causal-past entropy $S(t)$ (mean over six seeds, 18 events) for three sprout probabilities. Entropy is monotone in every run and ordered by the sprout fraction at every time: sprouts create causally independent events (parallelism), fills create dependencies (sequentialization).

3. **No visible deceleration:** growth is roughly linear at this scale, as expected, since 18 events barely populates tower level $d = 3$, so the mechanism of Section 3.7 cannot yet bite. **[GAP: sampling-based estimator for Ext (e.g. a Karzanov–Khachiyan-style Markov chain) to reach 10^2 – 10^3 events, plus implementation of $d \geq 3$ horn filling, then test for power-law $N(t)$ and for entropy deceleration]**

5 Discussion

5.1 The design space

Every framework paper hides choices. This section exposes ours, records the alternatives, and states why we chose as we did, so that each rejected branch remains a well-posed future project.

D1: What the cells are. Chosen: simplicial sets with directed edges. Alternatives: (a) bare undirected graphs, the simplest option, but no home for composition or coherence data; (b) oriented simplicial complexes with group-labelled edges, the minimal setting where torsion-type invariants (Reidemeister/Whitehead) are nonzero, attractive because torsion is precisely the information surviving below homotopy equivalence, rejected for v1 on grounds of economy; (c) Kan complexes as states, rejected because a Kan complex is a *finished* object (every horn filled): it is the natural *endpoint* of the dynamics, not its carrier.

D2: Which horns may be filled. Chosen: inner horns only; the limit object is a quasi-category. Alternative: all horns (Kan limit), giving an undirected, fully invertible structure. Rejected because directedness provides native causal orientation, keeps inverses costly rather than free, and makes orientation-reversal (the Feynman–Stückelberg reading of antimatter) expressible. The framework carries two arrows and they behave differently under this choice. The non-deletion arrow survives any filling policy, since a run never runs backwards, whereas the orientation arrow does not, which is the sharper cost of filling every horn.

D2': The groupoid completion. A variant places the mirror not beside the universe but on top of it, so that the same vertices carry the reverse of every edge and the completed object is a Kan complex, the standard model of an ∞ -groupoid. It delivers a total object that is trivial in the strongest sense. It carries three costs. Reversal exchanges the two outer horns and sends inner horns to inner horns, so no assignment of filling policies to branches produces it, and it can be had only by admitting outer horns as moves, which reverses D2. The two universes cease to be places, since they share every cell, and the two-branch picture with a big bang between them is lost. An object carrying all of its inverses is isomorphic to its own reversal at every stage, so the orientation arrow never appears at all. A further option avoids the last cost by making reversibility purchasable rather than built in, so that an outer-horn fill is an ordinary move at an ordinary price. Most edges then lack an inverse at any finite time, the orientation arrow exists as defined, and the Kan complex is the limit the process converges to, at which point heat death and full reversibility coincide. This variant is uncosted and remains open.

D3: The creation move. Chosen: sprouting from existing vertices. Alternatives: (a) a rich initial seed, rejected as engineered; (b) free appearance of disconnected cells, rejected as it breaks locality and connectedness. Sprouting preserves connectedness and the homotopy type (elementary expansion) by construction.

D4: The trigger rule. Chosen: none; the rule is to be derived by maximum entropy (Axiom 4, Conjecture 3.19). The postulated alternatives, each a different cosmology, are recorded for comparison: free sprouting (exponential, pure cosmological constant); frustration-triggered (self-limiting); fill-triggered (linear, expansion locked to the completion rate); curvature-coupled via discrete Ricci curvature (geometry back-reaction, Einstein-equation-shaped). Under the max-ent programme these should reappear as consequences of which constraints are imposed.

D4': One budget or many. Chosen: undecided, and the choice is recorded here because it fixes the growth law (Section 3.7). A single budget for the whole universe gives a move rate that saturates and a scale factor that decelerates. A budget per region gives activity proportional to the number of vertices and exponential growth. The locality constraint (C3) of Definition 3.18 favours the per-region reading, which would place a de Sitter phase in the late universe, but the constraint has not been formulated carefully enough to settle it.

D5: What entropy counts. Chosen (v1): orderings only, $S = \log \text{Ext}(J^-(e))$. Alternative: orderings *plus* filler choices, giving a two-component entropy whose second term measures distance from nervehood, i.e. exactly the quasicategory-vs-nerve gap where we have argued the physics lives. Deferred, not rejected: the ordering entropy is the right first theorem target, and the choice term requires the multiple-filler convention of Definition 3.3 to be settled first.

5.2 Open problems

In decreasing order of severity.

O1: The Lorentz group. Neither time dilation nor relativity of simultaneity is established. Proper time is defined as a longest chain (Definition 3.15), for which the Lorentz factor is a theorem whenever the causal order resembles a sprinkling into a Lorentzian manifold, so dilation reduces to the question of whether the orders this dynamics generates are of that kind. The required property is causal invariance: confluence of the rewriting system, so that all linearizations generate the same causal order and no experiment detects a preferred one. Whether confluence plus causal structure yields the actual Lorentz group is open here as it is in the hypergraph-rewriting programme [14]. A specific sub-problem: the framework predicts a fact of the matter

about the causal past of every event but must *not* predict a fact of the matter about “the total entropy of the universe now”; our slice-free entropy (Lemma 3.9) is designed for exactly this, but the claim needs a theorem, not a design intention.

O2: The area law. The long-game target is Jacobson’s derivation [16]: if the path entropy of a causal horizon is proportional to its *area*, the Einstein equations follow as an equation of state. If it is proportional to volume, the framework is dead as a theory of gravity. Computing the scaling of S across a causal horizon in the toy model is therefore the single most falsifying computation available to us. **[GAP: define “horizon” and “area” internally first]**

O3: Emergent unitarity. The substrate is irreversible; the effective dynamics encoded in the record must nevertheless be unitary and CPT-symmetric. The ledger-of-reversible-moves picture is a metaphor until an effective Hilbert-space dynamics is extracted; the difficulty is of the same kind as in the cellular-automaton interpretation of quantum mechanics.

O4: Geometry. Combinatorial volume is not metric geometry. Distances, dimension (spectral or Hausdorff), and curvature must be shown to emerge; discrete Ricci curvatures and spectral-dimension flow are the natural instruments, and comparison with the CDT dimensional-reduction results [3] is an obvious benchmark. Two constraints narrow the search. Coherence escalation creates cells of unbounded dimension, so by Remark 3.1 the emergent geometry must be built from the one-skeleton and the causal order alone, with higher cells contributing nothing metric. And by Remark 3.12 the weighted distance of Definition 3.11 is carried entirely by sprouts, so any emergent scale factor is a statement about vertex count rather than about cell count.

O5: The conserved charge of the tower. Filling promotes questions up the coherence tower monotonically. Whether a genuine conserved flux (“question current”) exists, and whether the two arrows (S and d) are provably aligned, is open.

5.3 Conclusion

We have proposed a framework in which the fundamental object is a growing directed simplicial set, time is the process of completing its inner horns, entropy is the count of forgotten orderings of a frame-invariant causal past, and the dynamical law is an inference rather than a postulate. The proved core is small, one experiment exists, and the failure modes are stated. The next steps are the two lemma proofs, the maximum-entropy derivation, a sampling estimator for long runs, and the horizon and area definitions that problem O2 requires.

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A The budget triangle

Let the move rate of any subsystem be c (Axiom 3) and suppose a subsystem translates at speed v . Writing the split as orthogonal components, the internal rate r satisfies $r^2 + v^2 = c^2$, hence

$$r = c \sqrt{1 - v^2/c^2},$$

the standard dilation factor, with the photon limit $r = 0$ at $v = c$.

The orthogonal composition is an assumption and not a consequence of the move set. Counting moves gives a linear split instead. A track of N events of which k are internal has $v/c = (N - k)/N$ and internal rate k/N , so that $r = c(1 - v/c)$, which differs from the expression above by a factor of three at $v = 0.8c$. Recovering the quadratic form requires the emergent geometry of linearizations to be Lorentzian rather than Euclidean, which is to say that straight worldlines must maximise accumulated internal filling. That is the content of problem O1, and Definition 3.15 is framed so that the required statement is the known longest-chain theorem for sprinklings rather than a new assumption.

B Simulation code

The toy implementation (also shipped as `bin/toy_universe.py`): sprout and Λ_1^2 -fill dynamics, event poset construction per Definition 2.9, and exact linear-extension counting by dynamic programming over downsets.

```
class Universe:
    # vertices, directed edges, unfilled inner horns;
    # every move is an event with explicit parent events.

    def sprout(self):
        # new vertex w, new edge v -> w
        # parents = { creator_event(v) }
        ...

    def fill(self, horn):
        # horn = (e1: u -> v, e2: v -> w)
        # new composite edge u -> w (+ 2-cell)
        # parents = { creator(e1), creator(e2) }
        ...

    def count_linear_extensions(parents):
        # DP over downsets of the event poset (bitmask),
        # processing masks in order of popcount.
        ...
```

Full listing in the repository. Runs: 18 events, six seeds, $p_{\text{sprout}} \in \{0.2, 0.5, 0.8\}$; output reproduced in Figure 3.